

1. A boy travels along a path given by $r = 5 \sin \theta$ where r is measured in feet. Then the distance in feet this boy traveled for $0 \leq \theta \leq 2\pi$ is $k\pi$. Determine the value of k .
2. An owl was perched on a vertical pine tree with his eyes 24 feet above ground. This owl spots a mouse on the level ground at an angle of depression of 50° . Determine the straight-line distance from the owl to the mouse. Express your answer rounded to the nearest tenth of a foot.
3. (k, w) is the solution to the matrix equation $\begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Determine the value of the sum $(k + w)$.
4. Let $k = 8 \left[\sum_{n=1}^{\infty} \left(\frac{1}{3} \right)^n \right]$. Determine the value of k .
5. Let $W(\theta) = (\cos \theta, \sin \theta)$ be the wrapping function around the unit circle. Determine the straight-line distance between $W(\pi)$ and $W\left(\frac{\pi}{4}\right)$. Express your answer as a decimal rounded to the nearest thousandth.
6. Two cruise ships leave the same port at the same time so that the angle between their paths is 60° . The first ship travels at 16 mph and the second ship travels at 14 mph. Determine how long they had been sailing at the time they are 26.424 miles apart. Express your answer as a decimal rounded to the nearest tenth of an hour.
7. Two lines $y_1 = \frac{1}{4}x - 3$ and $y_2 = 5x + \frac{1}{2}$ intersect. Determine the degree measure of the obtuse angle formed by this intersection. Express your answer as a decimal rounded to the nearest tenth of a degree.